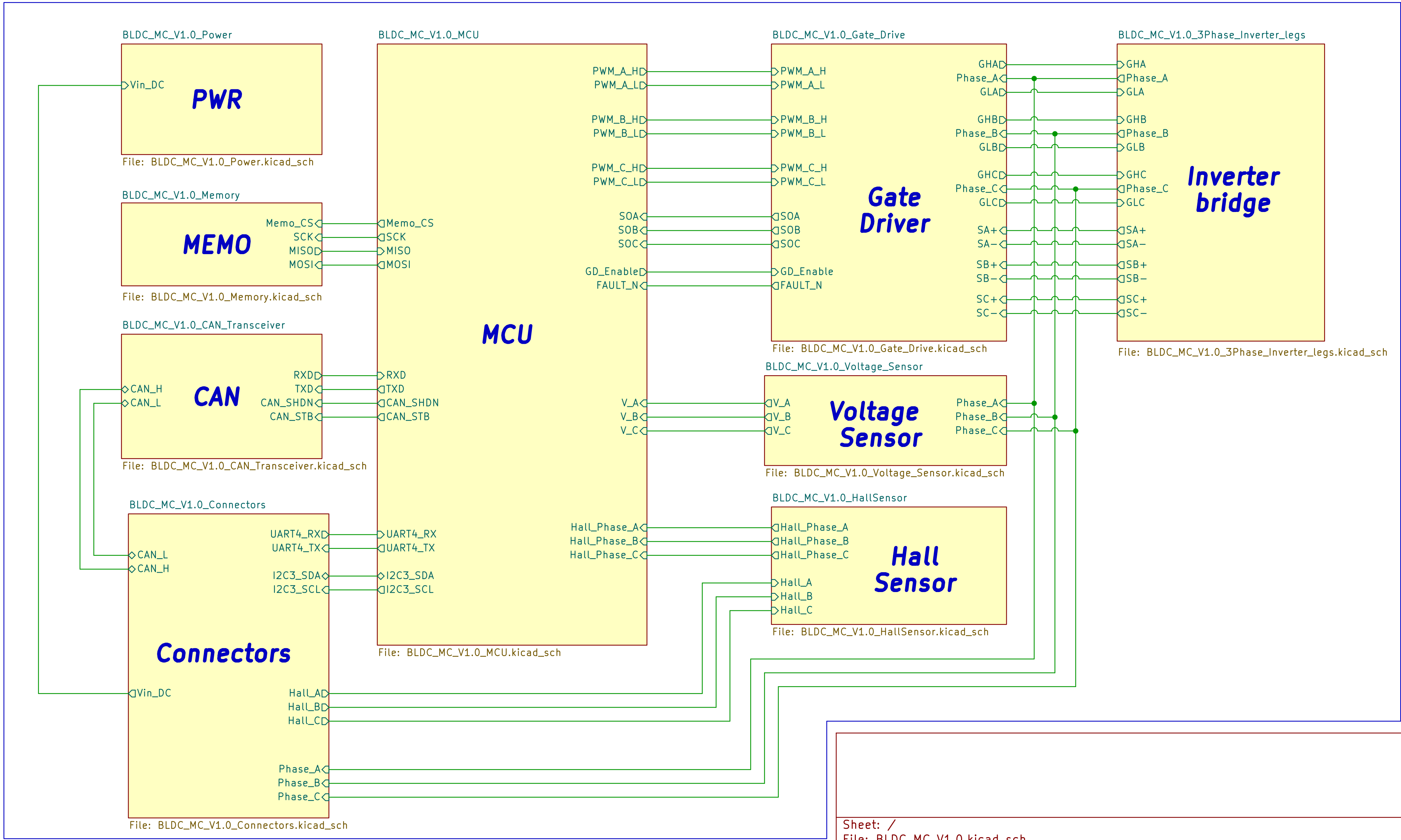
[1] BLDC_MC V1 60V 30A



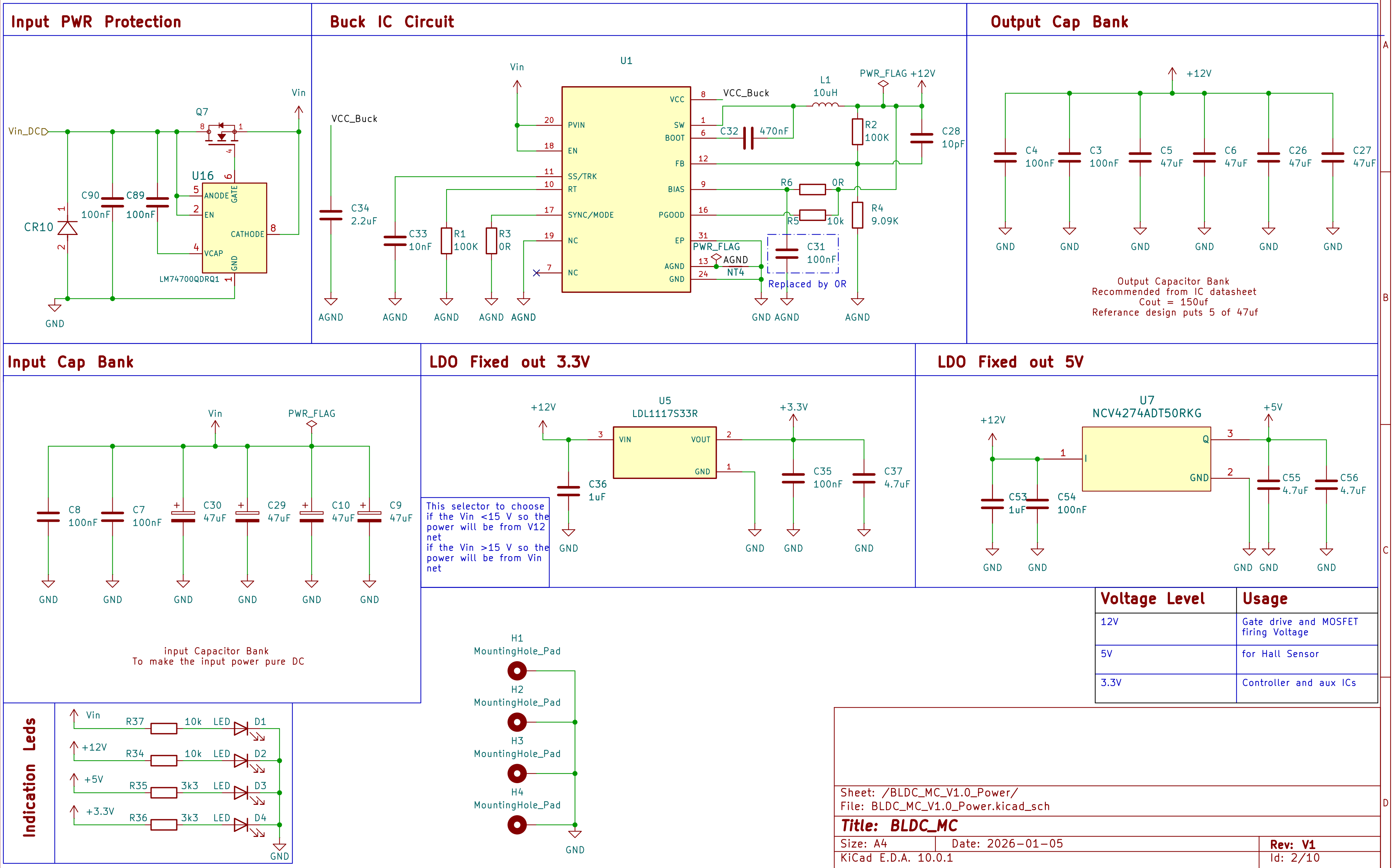
Sheet: /
File: BLDC_MC_V1.0.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-01-05	**Rev: V1**
KiCad E.D.A. 10.0.1		Id: 1/10

[1] Power Conversion Vin to 12V and 12V to 3.3V

The maximum input votage of this 60V so we can use for BLDC motors up to 60V



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

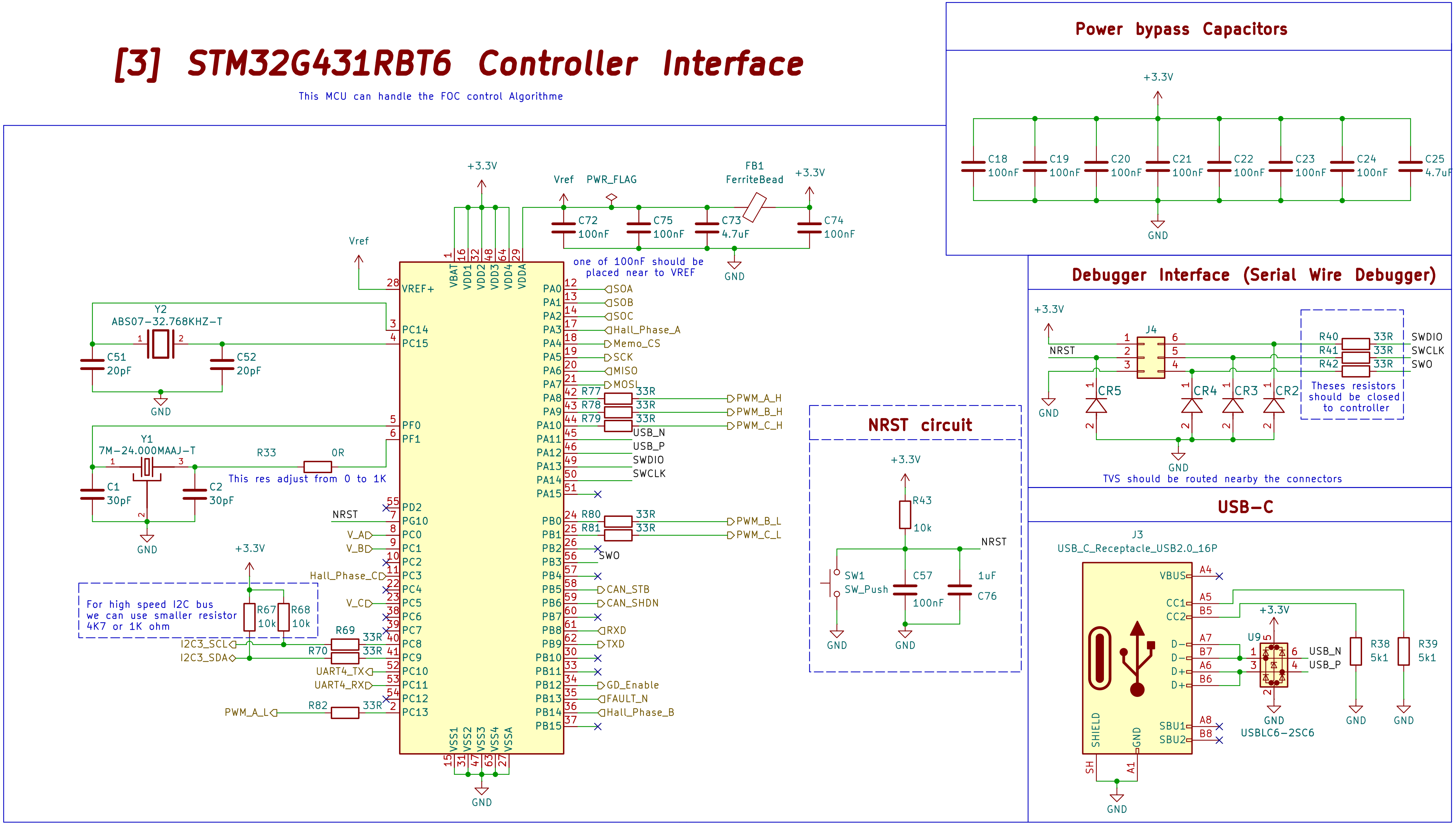
Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-05 **Rev: V1**
KiCad E.D.A. 10.0.1 Id: 2/10

[3] STM32G431RBT6 Controller Interface

This MCU can handle the FOC control Algorithme



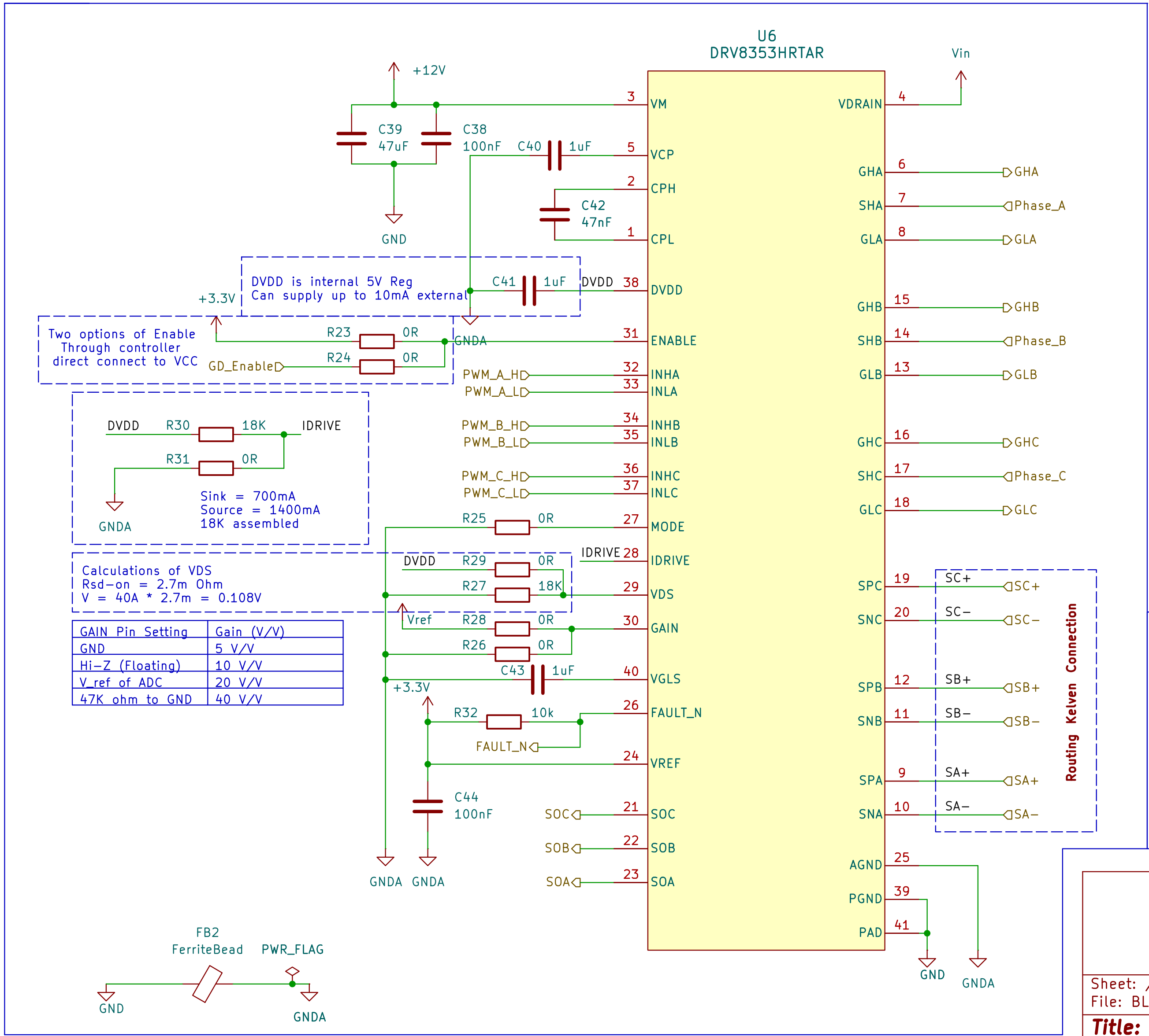
Sheet: /BLDC_MC_V1.0_MCU/
File: BLDC_MC_V1.0_MCU.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-03-23	Rev: V1
KiCad E.D.A. 10.0.1		Id: 3/10

[4] Smart Gate Drive

for the 3 phase inverter with current sensor



GAIN Pin Setting	Gain (V/V)
GND	5 V/V
Hi−Z (Floating)	10 V/V
V_ref of ADC	20 V/V
47K ohm to GND	40 V/V

Gate Resistors vs. IDRIVENormal Driver:

Uses external gate resistors to limit the current charging the MOSFET gate.
If you want to change the switching speed to reduce EMI or heat, you have to desolder and change the resistors.

Smart Driver: Uses an internal, programmable current source called IDRIVE. You set the source/sink current (e.g., 100mA, 1A) via a single pin or SPI. This allows you to tune the switching speed of your inverter without changing any hardware components.

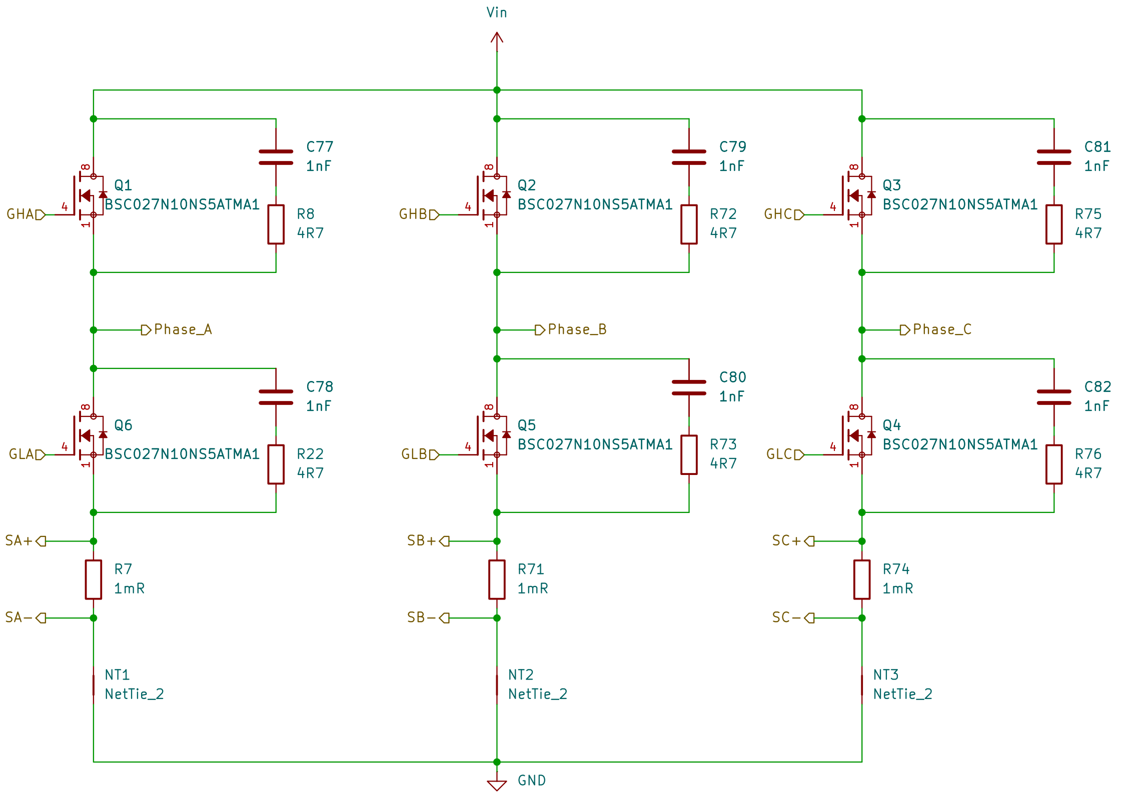
Sheet: /BLDC_MC_V1.0_Gate_Drive/
File: BLDC_MC_V1.0_Gate_Drive.kicad_sch

Title: BLDC_MC

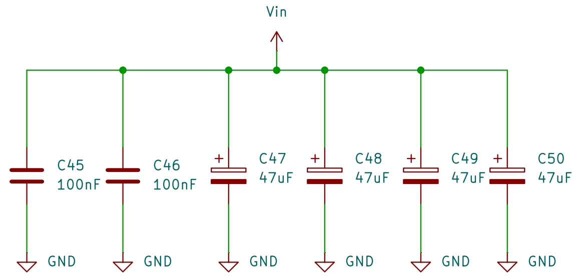
Size: A4 | Date: 2026−01−12 | **Rev: V1**
KiCad E.D.A. 10.0.1 | Id: 4/10

[5] 3 Phases Inverter Bridge

3 Phase Inverter Bridge



DC Link



Steps for adjusting snubber circuit

1− Measure the Initial Ringing
2− Add the "Shift" Capacitor −
increase the value of C_add until the ringing
frequency is exactly half of the original frequency f_new = 0.5 * f_ring).
3− Calculte the parasitic capacitance C_p = C_add / 3
4− Calculate Parasitic Inductance L_p −> L_p = 1/((2*pi*f_ring)^2 * C_p)
5− Determine the Snubber Resistor R_snub −> R_snub = sqrt{L_p/C_p}

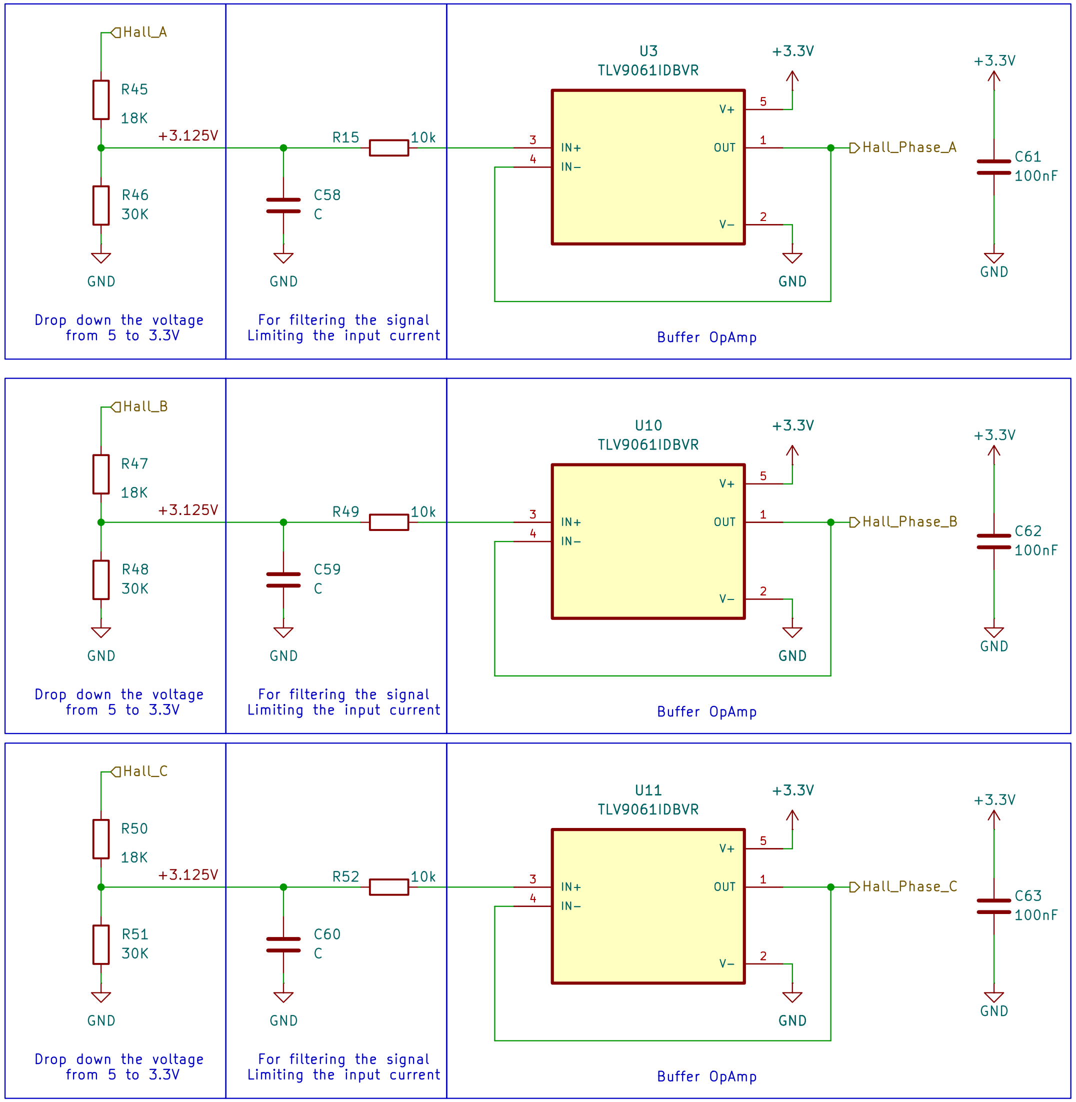
Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026−01−07	**Rev: V1**
KiCad E.D.A. 10.0.1		Id: 5/10

[6] Hall Effect Sensor Interface

For the 3 Phases of Hall Sensor

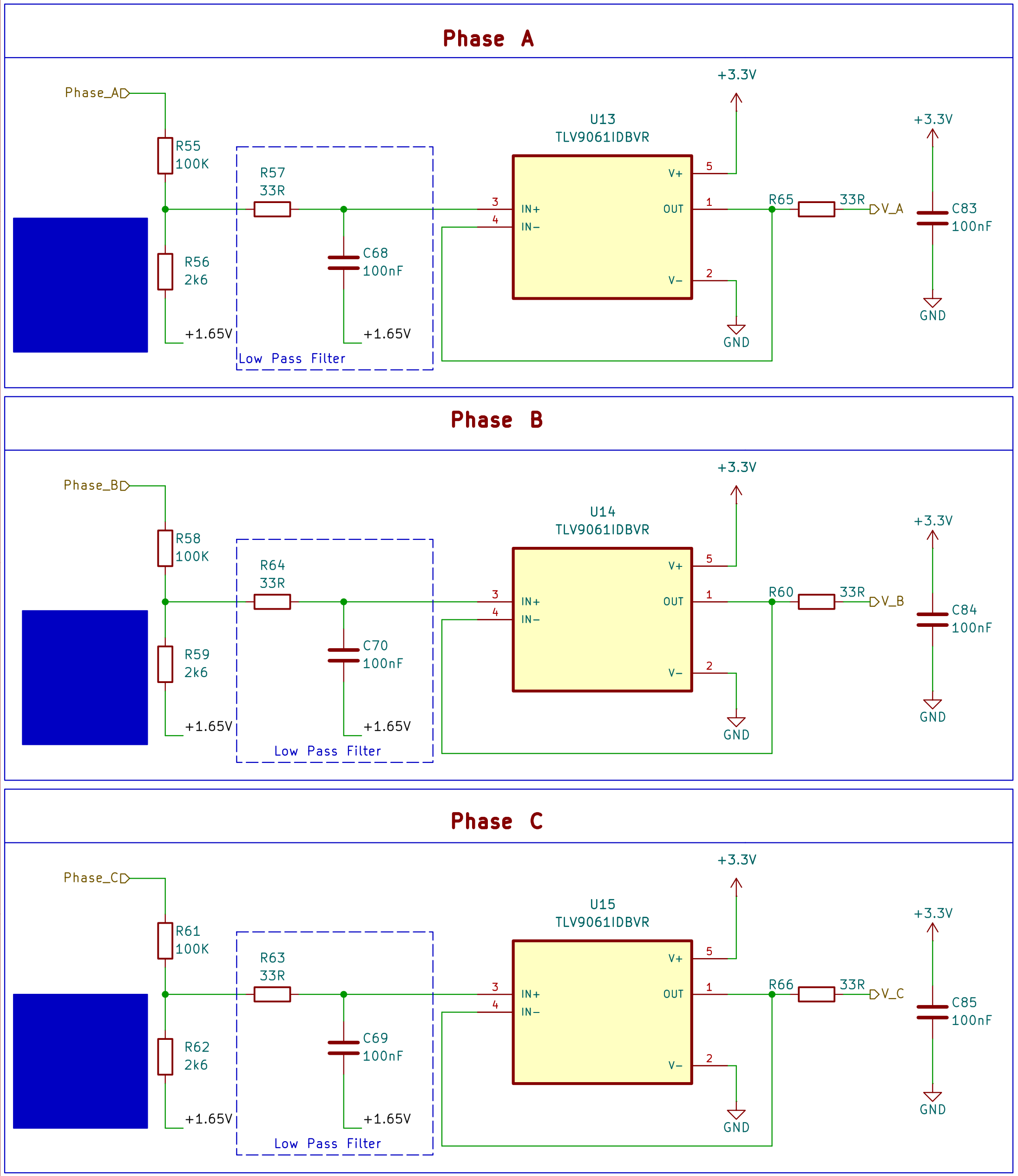


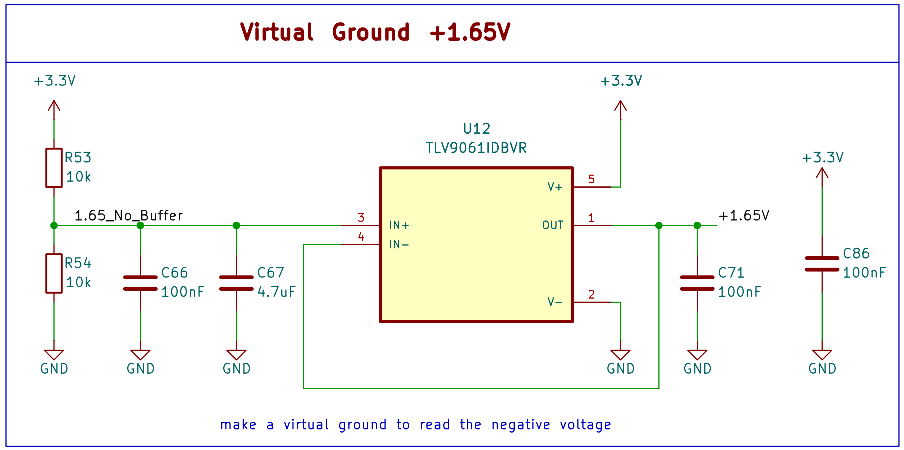
Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-03-05	**Rev: V1**
KiCad E.D.A. 10.0.1		Id: 6/10

[7] 3 Phase Inverter Voltage Sensing





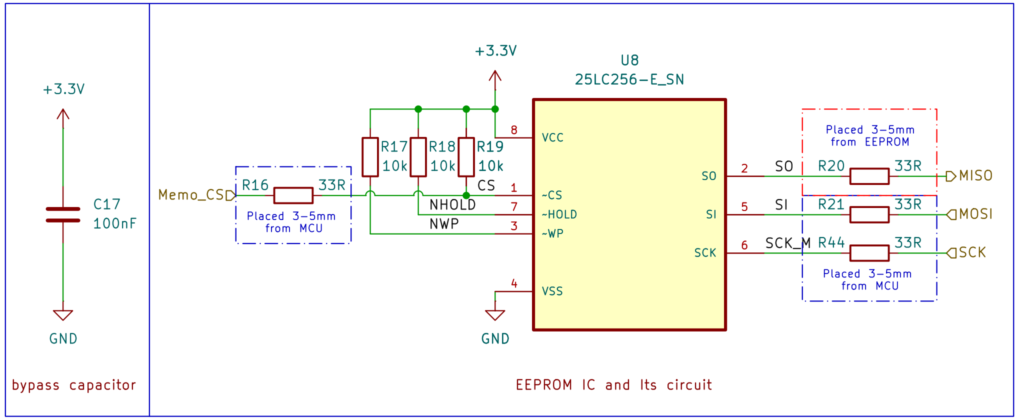
Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-03-11	Rev: V1
KiCad E.D.A. 10.0.1		Id: 7/10

[8] EEPROM Memory 256Kbit

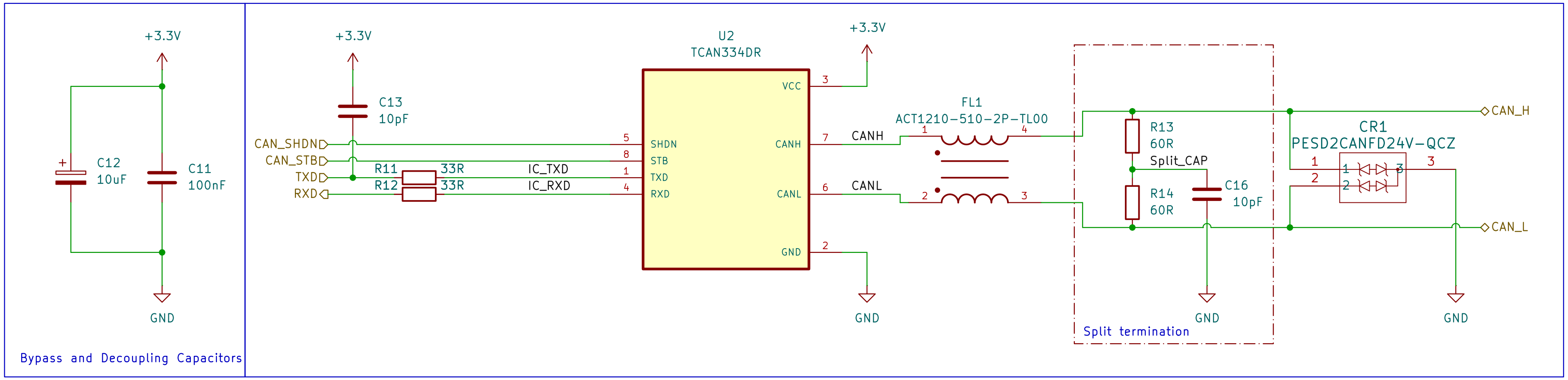
This memory used to store important data and configration parameters



Sheet: /BLDC_MC_V1.0_Memory/	
File: BLDC_MC_V1.0_Memory.kicad_sch	
Title: BLDC_MC	
Size: A4 Date: 2026-01-07	**Rev: V1**
KiCad E.D.A. 10.0.1	Id: 8/10

[9] CAN Transiver Classic CAN 1Mb/S

This IC can operate as CAN-FD for faster datarate



Note

Series Resistors Help manage signal integrity
by damping potential high-frequency noise or reflections
on the digital traces.

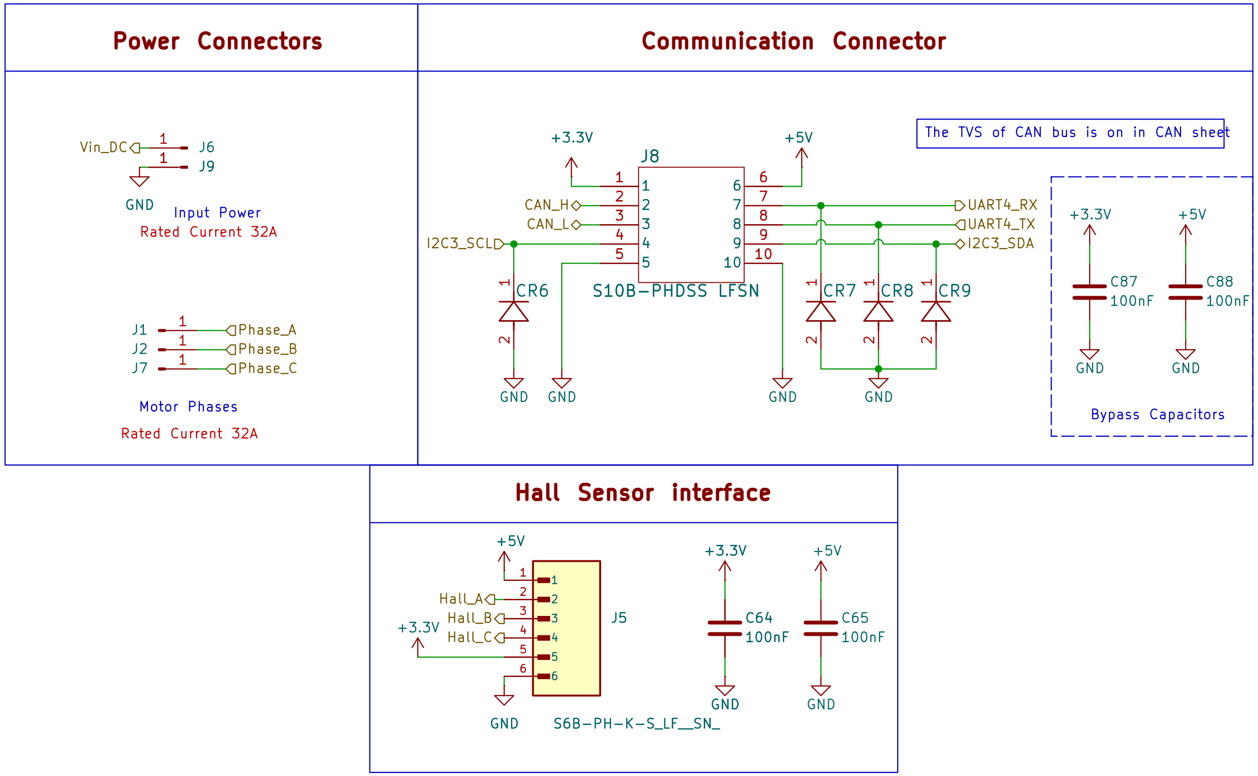
Parallel Capacitors is used to filter noise on the digital I/O lines

Sheet: /BLDC_MC_V1.0_CAN_Transceiver/
File: BLDC_MC_V1.0_CAN_Transceiver.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-01-05	Rev: V1
KiCad E.D.A. 10.0.1		Id: 9/10

[10] Board's Connectors Power & Communication



Sheet: /BLDC_MC_V1.0_Connectors/
File: BLDC_MC_V1.0_Connectors.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-03-08	**Rev: V1**
KiCad E.D.A. 10.0.1		Id: 10/10